Jin Miao

CONTACT INFORMATION

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ACADEMIC APPOINTMENT

Singapore Management University, Singapore — **July 2026 - Current**
Assistant Professor of Marketing

EDUCATION

University of Texas at Dallas, Richardson, TX — **Aug 2020 - May 2026**
Ph.D. Quantitative Marketing

Columbia University, New York City, NY — **Aug 2017 - May 2018**
M.S. Marketing Science

Tsinghua University, Beijing, China — **Aug 2013 - July 2017**
B.A. Economics & B.S. Psychology

RESEARCH INTERESTS

Substantive: Generative AI, Behavioral Economics
Methodology: Deep Learning / AI, Game Theory

PUBLICATION

Designing Loot Boxes: Implications for Profits and Welfare
Jin Miao, Sanjay Jain
Marketing Science (2024) vol. 43, no. 6, pp. 1242–1259.

Abstract: A loot box is a probabilistic allocation of virtual products, the exact outcome of which is known to consumers only after purchase. Consumers sometimes purchase these goods multiple times until their preferred products are obtained. As loot boxes have been gaining enormous popularity in recent years, they are often criticized as exploitative and socially wasteful. In this study, we develop a stylized model to study the optimal design of loot boxes and its impact on profits and social welfare. We find that firms may assign asymmetric probabilities to *ex ante* symmetric products. Firms could use loot boxes to offer products at low prices to users who would not buy these products under the traditional pricing strategy. Loot boxes enable firms to earn higher profits due to better price discrimination and market expansion. Contrary to the widespread criticism of loot boxes as socially harmful, our analysis reveals that the loot box strategy can improve social welfare. Some platforms promise that consumers can obtain their preferred products with no more than a certain number of purchases. Contrary to conventional wisdom, our analysis reveals that such a strategy can increase firm's profits while reducing consumer welfare.

WORKING PAPERS

Availability Bias and Design of Subscription Services: Pricing, Advertising and Welfare
Sanjay Jain, Jin Miao
under review

Abstract: Firms often sell subscriptions for services that consumers may or may not need. such as extended warranties, repair plans, and memberships for golf or the symphony. To choose a plan, a consumer must estimate how likely she is to need the service. Prior research on service pricing largely assumes that this estimate is unbiased. Empirical evidence shows that this is often not the case and that some consumers over-estimate their needs while others under-estimate them. We use the availability heuristic of Tversky and Kahneman (1973) to model how consumers form these estimates from memory. We study how a firm should design service contracts for such consumers. We find that the firm subsidizes consumers who over-estimate their needs by pricing below marginal

cost at the time of service. Furthermore, for rarely needed services the firm offers consumers a pure subscription with no charge at time of service. Contrary to intuition, we find that a small degree of bias benefits consumers and hurts the firm. We also show that the firm has an incentive to advertise, not to inform or persuade, but to make past needs salient. Such advertising raises prices even for consumers who are not biased. When the bias is strong, the firm distorts service quality upward and may offer enhancements that cost more than they are worth. When memory itself over-represents past needs, profits rise and the firm may subsidize all the subscribers. Finally, we find that perfect competition does not eliminate the negative impact of availability bias on social welfare, and some consumers can receive negative surplus even under perfect competition.

A Generative Model of Consumer Decision Dynamics in Limited-Time Products
Jin Miao, Fanglin Chen, Ying Xie

Teaching Experience

Marketing (Undergraduate Core), Singapore Management University
solo instructor — Fall 2026

Principles of Marketing (Undergraduate Core), the University of Texas at Dallas
solo instructor — Fall 2023, Fall 2024, Spring 2026
Outstanding Ph.D. Teacher Award (2025)

Invited Talks

Tsinghua University	October 2025
Chinese University of Hong Kong, Shenzhen	October 2025
University of Texas, Austin	October 2025
Singapore Management University	September 2025
City University of Hong Kong	September 2025

Grants

Tier-1 Research Grant, Singapore Ministry of Education	November 2026
Start-up Research Grant, Singapore Management University	July 2026
Doctoral Dissertation Early-Stage Grant, INFORMS Society of Marketing Science	Spring 2025

Honors Awards Scholarships

Graduation with High Distinction	Spring 2026
Nominee for President's Teaching Excellence Award	December 2025
Outstanding Ph.D. Teacher Award	Fall 2025
ISMS Doctoral Consortium Fellow	Summer 2025
AMA-Sheth Foundation Doctoral Consortium Fellow	Summer 2023
Betty and Gifford Johnson Travel Awards	Summer 2023

Conference Presentations

INFORMS Annual Conference	Atlanta GA, October 2025
INFORMS Marketing Science Annual Conference	Washington DC, June 2025
BizAI Annual Conference	Richardson TX, March 2025
INFORMS Marketing Science Annual Conference	Miami FL, June 2023
Production and Operations Management Conference	Orlando FL, May 2023

Last Update: October 2026